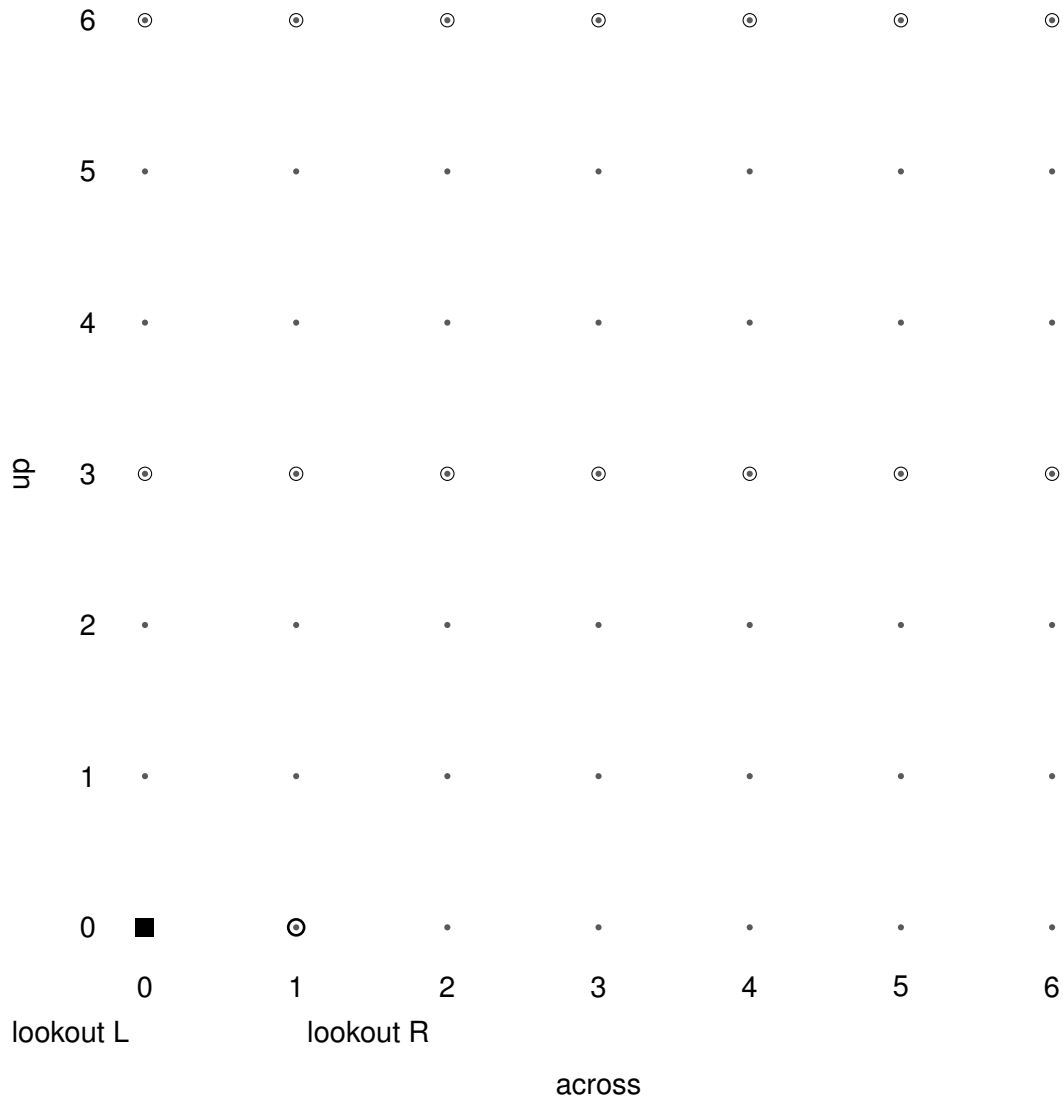


A target is hidden from a lookout if another dot lies exactly between them. Only dot centers count. Keep both lookouts fixed. The regularly spaced orchard continues beyond the printed grid.

Problem 1: Test every target on rows 3 and 6 from both lookouts. Mark B if both can see it, 1 if exactly one can see it, and 0 if neither can see it.

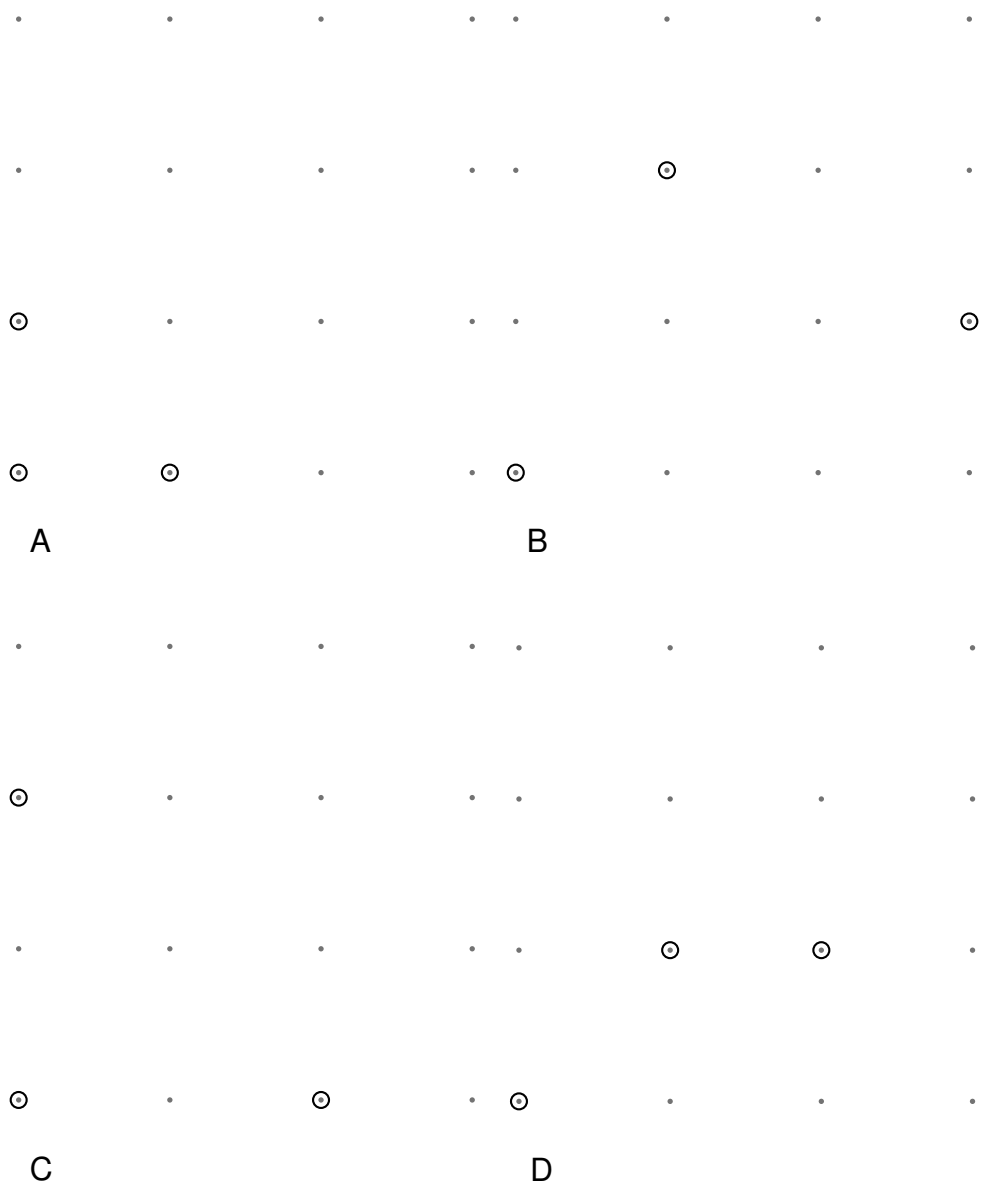


Across	0	1	2	3	4	5	6
Row 3							
Row 6							

Problem 2: Along row 4, can any target be hidden from both lookouts? You may go farther across than the printed grid. Find one or explain why none exists.

An empty triangle has no dots on its sides or inside except its three corners.

Problem 3: Make these four triangles with a ruler or rubber bands. Which are empty? Which have clear sides but dots inside? Mark every extra dot.



Problem 4: Make three empty triangles with different shapes on these grids. Find their areas in grid squares. Can an empty triangle have area larger than half a grid square? Explain your answer.

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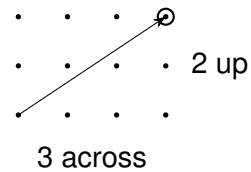
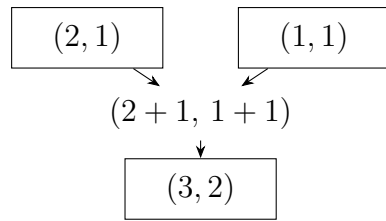
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A direction card gives *across*, *up*. To insert a card between two neighboring cards, add their across numbers and add their up numbers. Earlier cards stay in their order.



Problem 5: Start your ordered row with $(1, 0)$ and $(0, 1)$. Grow it until it contains both $(4, 3)$ and $(3, 4)$. Can the same rule ever produce $(4, 2)$? Explain your answer.

$(1, 0)$

$(0, 1)$

Problem 6: Your partner names a direction with positive across and up numbers no larger than 6, and no shared factor except 1. Can you always grow the row to reach it? Try several directions, then explain a way to decide where to insert cards.